TGDAnet: Text-Guided Domain Adaptation Network for Hyperspectral Binary Change Detection

Xuexiang Qin, Yuxiang Zhang, *Senior Member, IEEE*, Mingming Xu, *Senior Member, IEEE*, Wenxia Gan, *Member, IEEE*, and Yanni Dong, *Senior Member, IEEE*

***Abstract*—Deep-learning-based hyperspectral binary change detection (CD) methods have achieved impressive performance. However, their effectiveness is often constrained by the high cost of obtaining labeled hyperspectral images (HSIs). A natural and intuitive idea is to use knowledge learned from domains with rich labels and transfer it to unlabeled domains. Nevertheless, domain discrepancies significantly reduce detection performance when direct cross-domain transfer is applied. To address this challenge, we propose a text-guided domain adaptation network (TGDAnet) for hyperspectral binary CD. The proposed method leverages generalized and specialized domain text information to guide the model in extracting robust and transferable change features. Furthermore, a contrastive-learning-based strategy is incorporated to design global and local loss functions, thereby enhancing the sensitivity of the model to subtle and fine-grained changes across domains. The experimental results across four binary CD tasks on four hyperspectral scenes demonstrate that the proposed method outperforms existing state-of-the-art techniques. The code can be available at https://github.com/Pozei/TGDAnet**

***Index Terms*—Binary change detection (CD), contrastive learning, domain adaptation, hyperspectral image (HSI), text–vision model.**

I. Introduction

CHANGE detection (CD) aims to identify and analyze changes in a scene by comparing differences between multitemporal remotely sensed data [1], [2]. CD is widely used in various fields, such as urban management [3], environmental monitoring [4], and disaster assessment [5]. Hyperspectral images (HSIs) span a wide range of the electromagnetic spectrum, containing tens to hundreds of contiguous, high-resolution spectral dimensions [6], [7], [8], [9]. These properties make HSIs particularly well-suited for capturing subtle changes in CD tasks [10], [11].

Traditional HSIs-CD methods can be broadly classified into three categories: 1) algebra-based methods; 2) transform-based methods; and 3) classification-based methods.

1) Algebra-based methods identify change elements through algebraic analysis, with change vector analysis [12] and its variants [13], [14] being the most widely used.
2) Transform-based methods detect changes by projecting images into a feature space where interclass differences become more distinguishable. Among these, principal component analysis [15] and iteratively reweighted multivariate alteration detection [16] are the most representative.
3) Classification-based methods independently classify multitemporal images and then compare class differences across temporal phases to detect changes. Random forest [17] and support vector machine [18] are classical examples of such methods.

However, these methods exhibit inherent limitations: algebraic-based methods are vulnerable to radiometric variations, transform-based methods cannot adequately handle nonlinear changes, and classification-based methods depend heavily on label accuracy. These constraints force reliance on shallow features while failing to extract deep semantic information from HSIs, compromising CD accuracy and robustness.

Deep learning has been extensively applied to HSIs-CD tasks in recent years, owing to its powerful capability in feature extraction [19], [20], [21]. Convolutional neural networks (CNNs) are among the most commonly used architectures for HSIs-CD. Wang et al. [22] proposed a general end-to-end approach that enhanced the extraction of cross-channel gradient features by introducing a hybrid affinity matrix. Lu et al. [23] proposed a multitemporal symmetric fusion network for HSIs-CD, which used a two-branch CNN architecture to capture both the differential and temporal features. Graph convolutional networks (GCNs) have also demonstrated strong potential in HSIs-CD due to their ability to model complex spatial–spectral relationships. Yang et al. [24] developed a sharpening-aware graph network, which incorporated a novel Laplacian sharpening-based graph convolution

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Xuexiang Qin and Yanni Dong are with the School of Resource and Environmental Sciences, Wuhan University, Wuhan 430079, China (e-mail: qinxuexiang@whu.edu.cn; dongyanni@whu.edu.cn).

Yuxiang Zhang is with the School of Geophysics and Geomatics, China University of Geosciences, Wuhan 430074, China (e-mail: zhangyx@cug.edu.cn).

Mingming Xu is with the College of Oceanography and Space Informatics, China University of Petroleum (East China), Qingdao 266580, China (e-mail: xumingming@upc.edu.cn).

Wenxia Gan is with the School of Computer Science and Engineering, Wuhan Institute of Technology, Wuhan 430205, China (e-mail: ganwenxiagw@gmail.com).

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module to address the issue of disparity weakening caused by Laplacian smoothing and to enhance the representation of variations across bitemporal HSIs. Miao et al. [25] developed a cross-task meta-learning network with a graph-enhanced attention module, which designed a graph-enhanced attention module that leveraged the graph knowledge extracted from the graph transformer module to enhance the pixel-level feature representation. Transformer-based models have recently gained traction in HSIs-CD, benefiting from their ability to capture long-range dependencies. Wang et al. [26] proposed a joint spectral–spatial–temporal transformer for HSIs-CD. The method used positional encoding for each pixel within the data cube to retain spectral and spatial sequence information, while a spectral transformer encoder was used to extract spectral sequence features effectively. Zhang et al. [27] proposed a dual-branch Siamese spatial–spectral transformer attention network for HSIs-CD. The core idea was to fully leverage the complementary strengths of CNNs and transformers for comprehensive spatial and spectral feature extraction. In addition, Kolmogorov–Arnold networks (KANs) have also been explored in the context of HSIs-CD for their advantages in reducing model complexity, GPU memory usage, and computational overhead. Seydi et al. [28] showed that the KANs enhanced HSIs-CD performance without the extensive training requirements typical of deep networks. The Kolmogorov–Arnold theorem provided the theoretical foundation for this approach, offering a principled framework for hyperspectral data analysis by enabling effective dimensionality reduction and feature extraction from high-dimensional spectral signatures.

Although the aforementioned HSIs-CD methods have demonstrated impressive performance, several significant challenges warrant further attention, as outlined below:

1) Acquiring labeled HSIs are particularly challenging in the HSIs-CD tasks [27], [29]. The manual labeling of each scene is time-consuming, labor-intensive, and impractical in remote or inaccessible areas where ground-truth (GT) labels for HSIs cannot be obtained [30], [31], [32].
2) Most HSIs-CD methods primarily depend on vision cues and rarely on other knowledge to enhance vision representation learning. Text information can play a crucial role in enhancing vision representation learning within multimodal frameworks [33], [34].
3) Existing HSIs-CD methods primarily emphasize global-level feature extraction, often neglecting local structural details. This local-level feature plays a crucial role in identifying subtle information and thus contributes greatly to improving detection performance [35].

To effectively mitigate the aforementioned challenges, this article proposes a text-guided domain adaptation network (TGDAnet) for hyperspectral binary CD. The network enables CD in unlabeled target domains (TDs) through knowledge transfer from fully labeled source domains (SDs), leveraging cross-domain feature alignment to generalize detection capabilities. First, HSIs from both the domains are preprocessed to ensure consistent feature dimensions. Then, they are fed into a shared spatial–spectral transformer (SS-Transformer) to extract vision features, from which pseudolabels for the SD and TD are generated. Next, guided by the pseudolabels of both the SD and TD, the network constructs both the generalized and specialized domain text features. These text features are fused with the vision features to learn generalized cross-domain robustness representations. Finally, contrastive learning strategies are used to perform global and local domain adaptation for both the vision and vision–text features. The contributions of the method are summarized below:

1) *Domain Adaptation:* We adopt a domain adaptation strategy that enables effective knowledge transfer from the SD to TD to address the challenge of limited labeled data in HSIs-CD.
2) *Vision-Text Model:* We construct both the generalized and specialized text features for SD and TD, respectively. By fusing vision and text modalities, the model benefits from enhanced semantic guidance, resulting in more generalized and robust cross-domain CD features.
3) *Contrastive Learning:* We introduce global and local contrastive loss functions to better capture subtle and discriminative features. This design improves the sensitivity of the model to fine-grained variations across domains, thereby enhancing the overall effectiveness of CD.

The remainder of this article is structured as follows. Section II reviews related works. Section III details the proposed methodology. Section IV presents the experimental setup, results, and analysis. Finally, Section V concludes this article.

II. Related Works

A. Domain Adaptation

Due to the difficulty in data labeling and the reliance of deep learning on large amounts of annotated data, it remains challenging to further improve the accuracy of deep learning models. Domain adaptation addresses this challenge by transferring knowledge from a fully labeled SD to an unlabeled TD by reducing differences in the distribution of features between domains. Adversarial-learning-based domain adaptation has emerged as a significant research direction in the field of domain adaptation. Yu et al. [36] proposed a dynamic adversarial adaptation network, which dynamically and quantitatively assessed the relative significance of marginal and conditional distributions within adversarial learning. Domain adaptation based on the maximum mean discrepancy metric, which quantifies the distributional differences between SD and TD, is also a crucial research direction in this field. Wang et al. [37] introduced the dynamic distribution adaptation network, which used maximum mean discrepancy to measure the difference between marginal and conditional distributions across SD and TD, while A-distance dynamically evaluated their relative importance in the adaptation process. Although existing domain adaptation methods have shown effectiveness in HSIs' processing, their application to HSIs-CD remains underexplored. To effectively mitigate this problem, this article introduces domain adaptation techniques into the task of cross-domain HSIs-CD.

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B. Vision–Text Model

Contrastive language–image pretraining (CLIP) is a well-known model in text-supervised learning [38]. It used contrastive learning to align vision and text features in a shared space, enabling strong semantic connections between the two modalities [39]. Dong et al. [40] introduced a novel framework that leverages robust semantic information from image–text pairs, specifically designed for remote sensing CD. Their method reconstructed the original CLIP model to extract bitemporal features and incorporated a differential feature compensation module to capture fine-grained semantic changes between temporal observations. Jin et al. [41] proposed a language-guided dual-branch mamba network tailored for cross-scene HSI classification. In this approach, prior knowledge was incorporated through contrastive learning based on label and text-driven prior knowledge generation rules, thereby enhancing the model's capability to learn domain-invariant representations. Although vision–text fusion has achieved notable success in HSIs processing, its potential remains largely unexplored in the text of HSIs-CD. Therefore, this article proposes the incorporation of text information to guide the model for cross-domain CD.

C. Contrastive Learning

Contrastive learning is a self-supervised technique that focuses on learning informative representations by encouraging similar (positive) samples to be closer in the feature space while pushing dissimilar (negative) samples further apart, all without relying on labeled data. The information noise contrastive estimation loss $\mathcal{L}_{InfoNCE}$ stands out as one of the most widely used and influential loss functions in contrastive learning, playing a key role in many applications [42]

$$\mathcal{L}_{InfoNCE}(\mathbf{Q},\mathbf{K}) = -\frac{1}{b}\sum_{i=1}^{b}\log\left(\frac{\exp\left(\cos\left(q_i,k_i^+\right)/\tau\right)}{\sum_{j=1}^{b}\exp\left(\cos\left(q_i,k_j\right)/\tau\right)}\right). \tag{1}$$

Here, $\mathbf{Q}\in\mathbb{R}^{b\times d}$ is the query tensor and $\mathbf{K}\in\mathbb{R}^{b\times d}$ is the key tensor. b is the batch size. d is the feature dimension. q_i is a query vector for the ith sample. k_i^+ are positive keys paired with q_i. k_j represents all possible key vectors, including both the positive and negative samples. $\cos(\cdot,\cdot)$ is the cosine similarity function. τ is the temperature parameter. *log* is a cross-entropy operation on similarity scores. *exp* is an exponential function for logit scaling.

Chen et al. [43] proposed an improved prototypical network that integrated a convolutional block attention module-based feature enhancement algorithm with several widely used data augmentation strategies. Specifically, a supervised contrastive learning mechanism was introduced to address boundary ambiguity by pulling anchor samples closer to positive examples and pushing them farther from negative ones. In this framework, discriminative feature representations were used as constraints to guide prototype clustering, thereby improving classification accuracy and reducing model overfitting. Yao et al. [44] proposed a novel hypergraph contrastive learning network for large-scale HSI clustering. The network introduced a diffusion hypergraph-based contrastive clustering mechanism that constructed a diffusion hypergraph to capture high-order correlations in large-scale HSIs, thereby guiding contrastive learning to obtain more discriminative feature representations.

III. Proposed Method

A. Problem Definition

In this article, the domain comprises a labeled SD, $\mathbf{T}_s = \{(\mathbf{T}_s^1,\mathbf{T}_s^2),\mathbf{Y}_s\}$, and an unlabeled TD, $\mathbf{T}_t = \{(\mathbf{T}_t^1,\mathbf{T}_t^2)\}$. $\mathbf{T}_{s/t}^1$ and $\mathbf{T}_{s/t}^2$ represent a pair of temporally matched HSIs, where subscripts 1 and 2 correspond to images captured at two different time points. Each image is represented as $\mathbf{T}_{s/t}^{1/2}\in\mathbb{R}^{H_{s/t}\times W_{s/t}\times D_{s/t}}$, where $H_{s/t}$ and $W_{s/t}$ are the number of spatial dimensions, and $D_{s/t}$ is the number of spectral dimensions. To facilitate processing, the HSIs are divided into batches of size b and patches of size $p\times p$, which are reformatted as $\mathbf{T}_{s/t}^{1/2}\in\mathbb{R}^{b\times D_{s/t}\times p\times p}$. The GT change map $\mathbf{Y}_s\in\mathbb{R}^{H_s\times W_s}$ contains binary labels, where 0 indicates no change, and 1 indicates change. The objective of this work is to predict the binary change map $\mathbf{Y}_t\in\mathbb{R}^{H_t\times W_t}$ for the TD by transferring knowledge from SD. Fig. 1 illustrates the overall architecture of the proposed TGDAnet.

B. Vision Features' Extraction

1) Data Preprocessing: Due to the differences in spectral dimensions between the SD and TD, spectral dimensions' alignment preprocessing is essential to facilitate domain adaptation in HSIs-CD.

First, different features are computed by subtracting the bitemporal HSIs in the SD and TD. These features are then passed through the channel attention module (CAM) [45], which enhances the ability of the model to capture subtle changes by emphasizing informative spectral dimensions. Second, features from both the domains are mapped to a shared feature space of dimension D through a CNN consisting of a 3×3 2-D-CNN, a batch normalization, and a ReLU activation. Finally, the absolute values of the different features focus on the magnitude of the changes rather than their direction

$$\mathbf{T}_{s/t}^{dif} = CAM_{s/t}\left(\mathbf{T}_{s/t}^1 - \gamma_1\mathbf{T}_{s/t}^2\right) \tag{2}$$

$$D = (D_s + D_t)\,//num \tag{3}$$

$$\mathbf{T}_{s/t}^{dif} = \left\|CNN_{s/t}^{3\times3}\left(\mathbf{T}_{s/t}^{dif}\right)\right\|. \tag{4}$$

Here, γ_1 is a learnable scaling factor automatically optimized during training, and // denotes integration after division. Based on extensive experiments, the model achieves the best robustness when *num* is set to 9. Consequently, this value is fixed in subsequent experiments without additional hyperparameter optimization. $\|\cdot\|$ denotes the absolute value. The 3×3 CNN is used for channel dimension alignment, not only effectively maps and matches channel features but also captures local spatial structural information, thereby achieving a favorable balance between dimensional consistency and spatial representation capability.

We design an SS-Transformer as a basic backbone network, built upon the transformer architecture as the foundational backbone network to enhance the extraction of deep spatial

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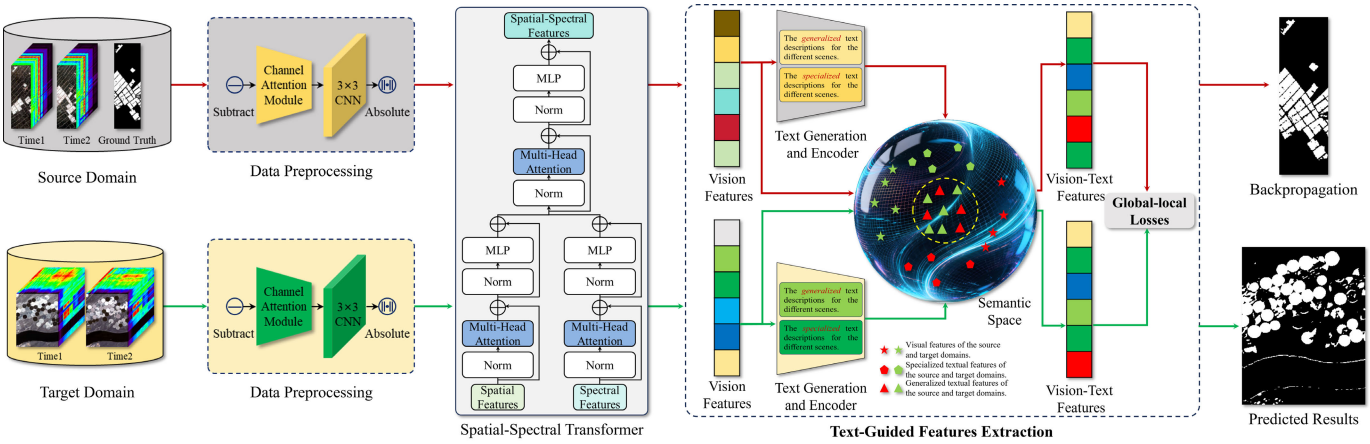


Fig. 1. Flowchart of the proposed TGDAnet.

and spectral features from HSIs. To promote effective domain adaptation, the same SS-Transformer architecture is used for both the SD and TD, ensuring consistent feature representation across domains.

2) Spatial Transformer: We reshape the different features $\mathbf{T}_{s/t}^{dif} \in \mathbb{R}^{b\times D\times p\times p}$ into spatial features $\mathbf{T}_{s/t}^{spa} \in \mathbb{R}^{b\times D\times p^2}$ and they are fed into the transformer

$$\mathbf{T}_{s/t}^{Spa} = Transformer_{spa}\left(\mathbf{PE}_{s/t}^{spa} + \gamma_2 \mathbf{T}_{s/t}^{spa}\right) \tag{5}$$

where $\mathbf{PE}_{s/t}^{spa} \in \mathbb{R}^{b\times D\times p^2}$ denotes the position encoding matrix, and γ_2 is a learnable scaling factor automatically optimized during training.

3) Spectral Transformer: We reshape the same $\mathbf{T}_{s/t}^{dif}$ into spectral features $\mathbb{R}^{b\times p^2\times D}$ for spectral sequence processing based on a transformer. The resulting spectral features $\mathbf{T}_{s/t}^{Spe} \in \mathbb{R}^{b\times(p^2+1)\times D}$ is reshaped and projected into $\mathbb{R}^{b\times D\times p^2}$ using a multilayer perceptron (MLP)

$$\mathbf{T}_{s/t}^{Spe} = Transformer_{spe}\left(cat\left(\mathbf{Z}_{s/t}^{spe}, \gamma_3 \mathbf{T}_{s/t}^{spe}\right)\right) \tag{6}$$

$$\mathbf{T}_{s/t}^{Spe} = MLP_{spe}\left(\mathbf{T}_{s/t}^{Spe}\right) \tag{7}$$

where $\mathbf{Z}_{s/t}^{spe} \in \mathbb{R}^{b\times 1\times D}$ denotes the class token in the respective token sequence, $cat(\cdot,\cdot)$ denotes that two tensors are concatenated, and γ_3 is a learnable scaling factor automatically optimized during training.

4) Spatial–Spectral Fusion Transformer: After obtaining both the spatial $\mathbf{T}_{s/t}^{Spa}$ and spectral $\mathbf{T}_{s/t}^{Spe}$ features, we concatenate them to get $\mathbf{T}_{s/t} \in \mathbb{R}^{b\times 2D\times p^2}$. Then, the features are permuted (PT) to $\mathbb{R}^{b\times p^2\times 2D}$ and fed into the transformer. Finally, the output features are projected into $\mathbb{R}^{b\times p^2\times d}$ using an MLP

$$\mathbf{T}_{s/t} = Transformer\left(PT\left(cat\left(\mathbf{T}_{s/t}^{Spe}, \gamma_4 \mathbf{T}_{s/t}^{Spa}\right)\right)\right) \tag{8}$$

$$\mathbf{T}_{s/t} = MLP\left(\mathbf{T}_{s/t}\right) \tag{9}$$

where γ_4 is a learnable scaling factor automatically optimized during training, and d is a hyperparameter empirically set to 512 in this work based on extensive experimental validation.

5) Postprocessing: The $\mathbf{T}_{s/t}$ are processed using a 2-D-CNN with a CNN of 7×7 and then reshaped to $\mathbf{T}_{s/t}^{vis} \in \mathbb{R}^{b\times(p^2\times d)}$. The final visual features $\mathbf{T}_{s/t}^{vis} \in \mathbb{R}^{b\times dim}$ are subsequently obtained through an MLP

$$\mathbf{T}_{s/t}^{vis} = MLP\left(CNN^{7\times 7}\left(\mathbf{T}_{s/t}\right)\right) \tag{10}$$

where *dim* is set to 512 to match the dimensionality of the text features, thereby facilitating effective fusion between vision and text modalities. The 7×7 CNN is introduced to expand the network's receptive field, enabling it to capture richer global contextual information.

6) Vision Domain Adaptation Loss: In this article, we design both the global and local loss functions based on (1) to enhance cross-domain subtle CD and improve model sensitivity to fine-grained variations

$$\mathcal{L}_{global}\left(\mathbf{Q},\mathbf{K}\right) = \frac{1}{2}\left(\mathcal{L}_{InfoNCE}\left(\mathbf{Q},\mathbf{K}\right) + \mathcal{L}_{InfoNCE}\left(\mathbf{K},\mathbf{Q}\right)\right) \tag{11}$$

$$\mathcal{L}_{local}\left(\mathbf{Q},\mathbf{Y}_q,\mathbf{K},\mathbf{Y}_k\right) = \sum_{i=c}^{C}\mathcal{L}_{global}\left(\mathbf{Q}^{(c)},\mathbf{K}^{(c)}\right) \tag{12}$$

where C denotes the number of classes, and $\mathbf{Y}_q$ and $\mathbf{Y}_k$ denote the labels of $\mathbf{Q}$ and $\mathbf{K}$, respectively. Equation (11) introduces a symmetric global contrastive loss based on (1), which enforces contrastive constraints in both the $\mathbf{Q}\rightarrow\mathbf{K}$ and $\mathbf{K}\rightarrow\mathbf{Q}$ directions, thereby promoting bidirectional alignment between input feature representations. This design improves the stability and consistency of cross-domain feature alignment. Building upon this formulation, (12) constructs multiple local components according to semantic categories and applies the global contrastive loss independently to the feature representations of each category. The contrastive constraints across all the categories are then aggregated to form the final local contrastive loss.

We generate pseudolabels for $\hat{\mathbf{Y}}_s$ and $\hat{\mathbf{Y}}_t$ using $\mathbf{T}_{s/t}^{vis} \in \mathbb{R}^{b\times dim}$ through an MLP and construct global and local vision domain adaptation loss $\mathcal{L}_{vda}$ based on (11) and (12) to minimize cross-domain distribution discrepancies

$$\mathcal{L}_{vda} = \mathcal{L}_{\text{global}}\left(\mathbf{T}_s^{vis},\mathbf{T}_t^{vis}\right) + \mathcal{L}_{\text{local}}\left(\mathbf{T}_s^{vis},\hat{\mathbf{Y}}_s,\mathbf{T}_t^{vis},\hat{\mathbf{Y}}_t\right). \tag{13}$$

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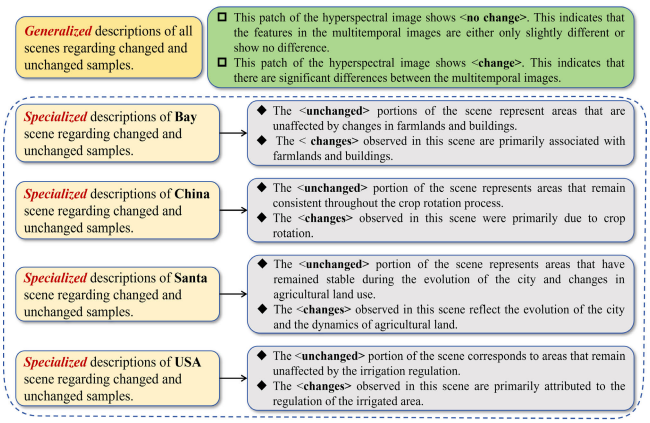


Fig. 2. Generalized and specialized text descriptions for the four scenes.

In this work, we propose a contrastive learning loss that synergistically integrates global and local interactions. At the global level, the loss imposes contrastive constraints to align the overall feature distributions between SD and TD. Simultaneously, it introduces local contrastive constraints that capture fine-grained correspondences between the two domains, thereby enabling more precise and refined cross-domain distribution alignment.

In particular, the initial pseudolabels may contain noise, causing the generated textual descriptions to deviate from the true semantic meaning. However, as the model performance improves through continuous optimization, the pseudolabels' accuracy increases, and the generated descriptions become more aligned with the actual semantic content.

C. Text-Guided Features' Extraction

Text contains rich contextual features, such as underlying causes of change, geographic location, and climatic environment, which contribute interpretive insights beyond visual cues for a more comprehensive CD.

1) Text Generation: First, the specialized text descriptions $\mathbf{T}_{s/t}^{spe}$ are generated from $\hat{\mathbf{Y}}_s$ and $\hat{\mathbf{Y}}_t$ to highlight the key causes of change in each scene. Second, the generalized text descriptions $\mathbf{T}_{s/t}^{gen}$ are generated from $\hat{\mathbf{Y}}_s$ and $\hat{\mathbf{Y}}_t$ to capture universal human perceptual judgments of change and nonchange. In this article, specialized features refer to those that characterize domain-specific attributes, whereas generalized features denote those with cross-domain representational capability and stronger generalization performance. Examples of text descriptions are illustrated in Fig. 2.

2) Text Encoding: To improve the overall efficiency of the text encoding process, we first apply the lowercase byte pair encoding method [46] to segment the input text. Subsequently, the key architectural components and hyperparameters of the text encoder from the pretrained ViT-B-32 model are adopted to construct the text encoding network. The resulting tokenized representations are then fed into a transformer encoder [47] for feature modeling, ultimately producing the text feature representation $\mathbf{T}_{s/t}^{gen/spe} \in \mathbb{R}^{b\times dim}$, $dim = 512$.

3) Specialized Vision–Text Features: We extract specialized domain vision–text features $\mathbf{X}_{s/t}^{spe}$ by computing the discrepancy between vision features and generalized text features. Based on these specialized vision–text features and their corresponding text representations, we formulate a specialized loss function $\mathcal{L}_{spe}$ based on (11)

$$\mathbf{X}_{s/t}^{spe} = MLP\left(\mathbf{T}_{s/t}^{vis} - \gamma_5 \mathbf{T}_{s/t}^{gen}\right) \tag{14}$$

$$\mathcal{L}_{spe} = \mathcal{L}_{global}\left(\mathbf{T}_{s}^{spe}, \mathbf{X}_{s}^{spe}\right) + \mathcal{L}_{global}\left(\mathbf{T}_{t}^{spe}, \mathbf{X}_{t}^{spe}\right) \tag{15}$$

where γ_5 is a learnable scaling factor automatically optimized during training.

4) Generalized Vision–Text Features: We extract generalized domain vision–text features $\mathbf{X}_{s/t}^{gen}$ by computing the discrepancy between vision features and specialized text features. Based on these generalized vision–text features and their corresponding text representations, we formulate a generalized loss function $\mathcal{L}_{gen}$ based on (11) and (12)

$$\mathbf{X}_{s/t}^{gen} = MLP\left(\mathbf{T}_{s/t}^{vis} - \gamma_6 \mathbf{T}_{s/t}^{spe}\right) \tag{16}$$

$$\mathcal{L}_{gen} = \mathcal{L}_{global}\left(\mathbf{X}_{s}^{gen}, \mathbf{X}_{t}^{gen}\right) + \mathcal{L}_{local}\left(\mathbf{X}_{s}^{gen}, \hat{\mathbf{Y}}_s, \mathbf{X}_{t}^{gen}, \hat{\mathbf{Y}}_t\right) \tag{17}$$

where γ_6 is a learnable scaling factor automatically optimized during training.

5) Vision–Text Features: We combine generalized vision–text features with generalized text features to derive deeper robust cross-domain representations $\mathbf{X}_{s/t}$, and introduce a vision–text domain adaptation loss $\mathcal{L}_{vtda}$ to minimize cross-domain distribution discrepancies

$$\mathbf{X}_{s/t} = \mathbf{X}_{s/t}^{gen} + \gamma_7 \mathbf{T}_{s/t}^{gen} \tag{18}$$

$$\mathcal{L}_{vtda} = \mathcal{L}_{global}\left(\mathbf{X}_s, \mathbf{X}_t\right) + \mathcal{L}_{local}\left(\mathbf{X}_s, \hat{\mathbf{Y}}_s, \mathbf{X}_t, \hat{\mathbf{Y}}_t\right) \tag{19}$$

where γ_7 is a learnable scaling factor automatically optimized during training.

D. Prediction

1) Total Loss Function: In HSIs-CD tasks, the problem is commonly formulated as a binary classification problem, where networks are commonly trained using the cross-entropy loss function. The cross-entropy loss $\mathcal{L}_{ce}$ is defined as

$$\mathcal{L}_{ce} = -\frac{1}{b}\sum_{i=1}^{b}\left(\mathbf{Y}_s^i \log \hat{\mathbf{Y}}_s^i + \left(1 - \mathbf{Y}_s^i\right)\log\left(1 - \hat{\mathbf{Y}}_s^i\right)\right) \tag{20}$$

where $\mathbf{Y}_s^i$ is the GT binary label of the ith sample, and $\hat{\mathbf{Y}}_s^i$ is the predicted label of the ith sample.

An effective loss function is critical for optimizing model architecture, enhancing feature discriminability, and improving prediction accuracy. In TGDAnet, we formulate the total loss function by integrating key components

$$\mathcal{L}_{loss} = \mathcal{L}_{ce} + \lambda_1\left(\mathcal{L}_{vda} + \mathcal{L}_{vtda}\right) + \lambda_2\left(\mathcal{L}_{spe} + \mathcal{L}_{gen}\right) \tag{21}$$

where λ_1 and λ_2 are assigned to the two hyperparameters.

2) Prediction: The network is optimized by minimizing $\mathcal{L}_{loss}$ to enhance its ability to detect significant changes. The optimized model is then transferred to the TD to predict the result $\hat{\mathbf{Y}}_t$.

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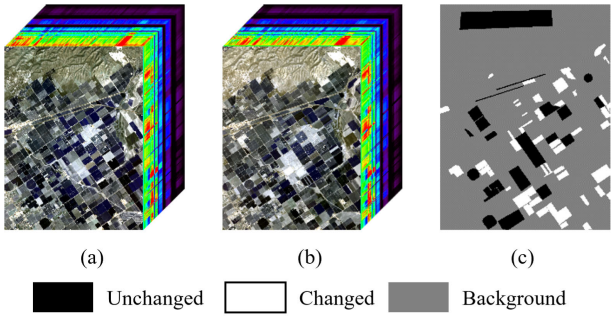


Fig. 3. Bitemporal HSIs (R: 10, G: 15, and B: 20) and GT for the Santa scene. (a) Image was acquired in 2013. (b) Image was acquired in 2014. (c) Image represents the GT.

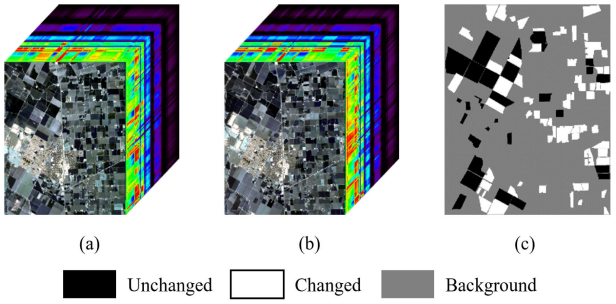


Fig. 4. Bitemporal HSIs (R: 10, G: 15, and B: 20) and GT for the Bay scene. (a) Image was acquired in 2013. (b) Image was acquired in 2015. (c) Image represents the GT.

IV. Experiments

A. Data Description

This work evaluates the effectiveness of TGDAnet using four hyperspectral scenes: Santa, Bay, USA, and China. The detailed description of each scene is given below:

1) *Santa:* The Santa scene was acquired by the airborne visible/infrared imaging spectrometer (AVIRIS) in 2013 and 2014, covering the city of Santa Barbara, CA, USA. The scene was acquired across the 0.4–2.5 μm spectral range with a nominal spatial resolution of 20 m. Each HSI consists of 224 spectral dimensions and has a spatial dimension of 984 $\times$ 740 pixels. The changes observed in this scene reflect the evolution of the city and the dynamics of agricultural land [48]. Fig. 3 presents a pseudocolor composite image (R: 10, G: 15, and B: 20) of the area along with the corresponding GT change map.
2) *Bay:* The Bay scene was acquired by the AVIRIS in 2013 and 2015, covering the city of Patterson, California, United States, and its surrounding areas. The scene was acquired across the 0.4–2.5 μm spectral range with a nominal spatial resolution of 20 m. Each HSI consists of 224 spectral dimensions and has a spatial dimension of 600 $\times$ 500 pixels. The changes observed in this scene are primarily associated with farmlands and buildings [49]. Fig. 4 presents a pseudocolor composite image (R: 10, G: 15, and B: 20) of the area along with the corresponding GT change map.
3) *USA:* The USA scene was collected on May 1, 2004, and May 8, 2007, over an irrigated agricultural field in the United States. It was collected using the Hyperion sensor aboard the EO-1 satellite, covering a wavelength range of 0.4–2.5 μm. The image features a spatial resolution of approximately 30 m and a spectral resolution of around 10 nm. Each HSI contains 154 spectral dimensions with a spatial size of 307 $\times$ 241 pixels. The changes observed in this scene are primarily attributed to the regulation of the irrigated area [50]. Fig. 5 displays the bitemporal false-color images (R: 10, G: 15, and B: 20) and the corresponding GT change maps.

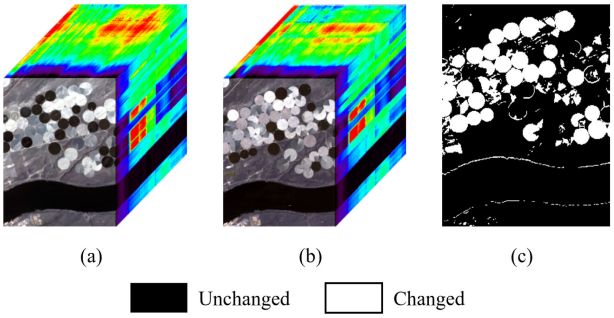


Fig. 5. Bitemporal HSIs (R: 10, G: 15, and B: 20) and GT for the USA scene. (a) Image was collected on May 3, 2013. (b) Image was collected on December 31, 2013. (c) Image represents the GT.

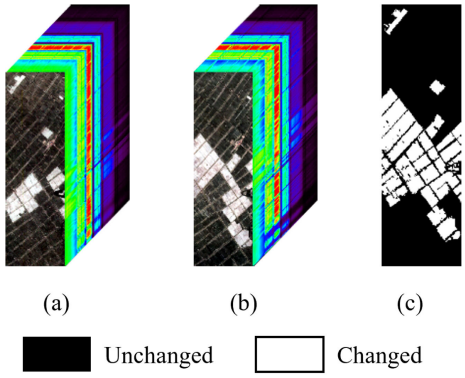


Fig. 6. Bitemporal HSIs (R: 10, G: 15, and B: 20) and GT for the China scene. (a) Image was collected on May 3, 2006. (b) Image was collected on April 23, 2007. (c) Image represents the GT.

4) *China:* The China scene was collected using the Hyperion sensor aboard the EO-1 satellite on May 3, 2006, and April 23, 2007. The area is located in Yancheng City, Jiangsu Province, China, primarily covering farmland regions. The images span a wavelength range of 0.4–2.5 μm, with a spatial resolution of approximately 30 m and a spectral resolution of 10 nm. Each HSI includes 155 spectral dimensions and has a spatial size of 450 $\times$ 140 pixels. The changes observed in this scene were primarily due to crop rotation [51]. Fig. 6 presents the bitemporal false-color composite images (R: 10, G: 15, and B: 20) along with their corresponding GT change maps.

B. Experimental Settings

To systematically verify the effectiveness of TGDAnet, we have designed four cross-domain adaptation tasks based on sensor similarity, covering four typical application scenarios,

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TABLE I
NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES AND LABELS FOR THE FOUR TASKS

Tasks		Training		Validation	Testing
		Source	Target	Source	Target
		2.5%	2.5%	10.0%	100.0%
Bay→Santa	Samples	1837	3313	7348	132552
	Labels	1837	0	7348	0
Santa→Bay	Samples	3313	1837	13255	73481
	Labels	3313	0	13255	0
China→USA	Samples	1575	1850	6300	73987
	Labels	1575	0	6300	0
USA→China	Samples	1850	1575	7399	63000
	Labels	1850	0	7399	0

to closely approximate the real remote sensing application conditions. Specifically: 1) "Bay→Santa"; 2) "Santa→Bay"; 3) "China→USA"; and 4) "USA→China." In each case, the symbol "→" on the left indicates SD and the one on the right indicates TD.

We use 11×11 image patches and a batch size of 256 to maintain parameter consistency and ensure fair comparisons. The model is trained for 50 epochs using the AdamW optimizer with a learning rate of 0.0001 and a weight decay of 0.8. In addition, the SS-Transformer architecture includes the transformer module with nine layers. The multihead attention mechanism uses six heads with a dimension of 16 each, while the feedforward network has an intermediate layer dimension of 5, and the dropout rate is set to 0.6.

For sample partitioning, 2.5% of the combined source-target samples are used for training, 10.0% of the source samples are used for validation, and the entire TD (100.0%) is used for testing. Detailed sample sizes for training, validation, and testing samples and labels are provided in Table I.

We use four standard metrics to quantitatively assess detection performance: intersection-over-union (IoU), precision (Prec), recall (Recall), and $F1$ score ($F1$). These are derived from the confusion matrix components

$$IoU = \frac{TP}{TP + FP + FN} \times 100\% \tag{22}$$

$$Prec = \frac{TP}{TP + FP} \tag{23}$$

$$Recall = \frac{TP}{TP + FN} \tag{24}$$

$$F1 = 2 \times \frac{Prec \times Recall}{Precision + Recall} \times 100\%. \tag{25}$$

Here, true positives (TPs) are correctly detected as changed pixels. False positives (FPs) are unchanged pixels falsely detected as changes. False negatives (FNs) are changed pixels missed by detection. True negatives (TNs) are correctly identified as unchanged pixels.

To evaluate the performance of TGDAnet, we have selected several classical and novel methods for comparison, including DAAN (2019) [36], DDAN (2020) [37], TSTnet (2021) [52], CMJAN (2024) [53], CACL (2024) [54], and S^{4}DL (2025) [55]. Specifically, DAAN, DDAN, and TSTnet are traditional methods for domain adaptation, while CACL and S^{4}DL are more recent methods in this field. On the other hand, CMJAN is a newer method designed specifically for CD in domain adaptation. In particular, we adopt a consistent vision data preprocessing procedure for domain adaptation algorithms not designed explicitly for HSIs-CD to ensure fair and comparable experimental results.

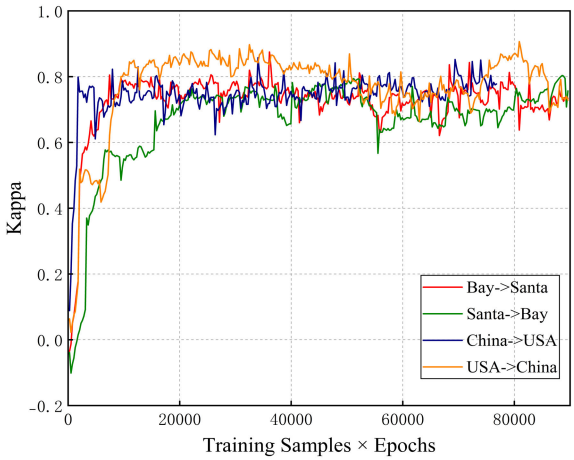


Fig. 7. Accuracy variation curve of pseudolabels for the TD during training.

In addition, Prec is the proportion of samples predicted as positive that are positive, while the Recall is the proportion of actual positive samples correctly identified by the model. However, these two metrics only provide a partial view of individual model performance and do not fully capture the overall effectiveness of each method. In contrast, the IoU measures the overlap between predictions and GT, while $F1$ balances precision and recall via their harmonic mean. IoU and $F1$ offer a more comprehensive evaluation of network performance. Therefore, we will place greater emphasis on analyzing the performance of TGDAnet in terms of IoU and $F1$ in Sections IV-D–IV-F. Finally, the results presented in this article are averaged over three independent experiments.

C. Analysis of Pseudolabel Accuracy

To better evaluate the impact of pseudolabels, we adopt the Kappa coefficient as the evaluation metric, which is defined as follows:

$$N = TP + TN + FP + FN \tag{26}$$

$$p_o = \frac{TP + TN}{N} \tag{27}$$

$$p_e = \frac{(TP + FP)(TP + FN) + (FN + TN)(FP + TN)}{N^2} \tag{28}$$

$$Kappa = \frac{p_o - p_e}{1 - p_e}. \tag{29}$$

As shown in Fig. 7, the accuracy of pseudolabels in the TD remains extremely low at the early stage of training, increases rapidly during the middle stage, and eventually stabilizes at a high level despite minor fluctuations in the later stage.

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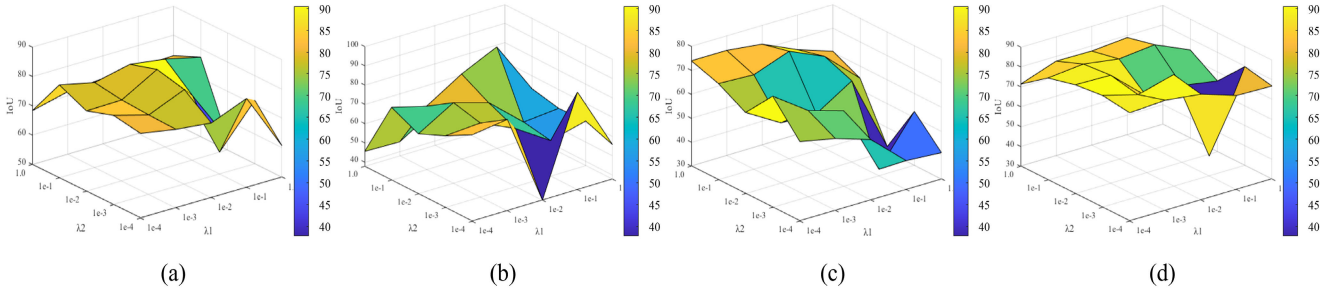


Fig. 8. IoU of TGDAnet with different hyperparameters λ_1 and λ_2 on four tasks. (a) Bay→Santa. (b) Santa→Bay. (c) China→USA. (d) USA→China.

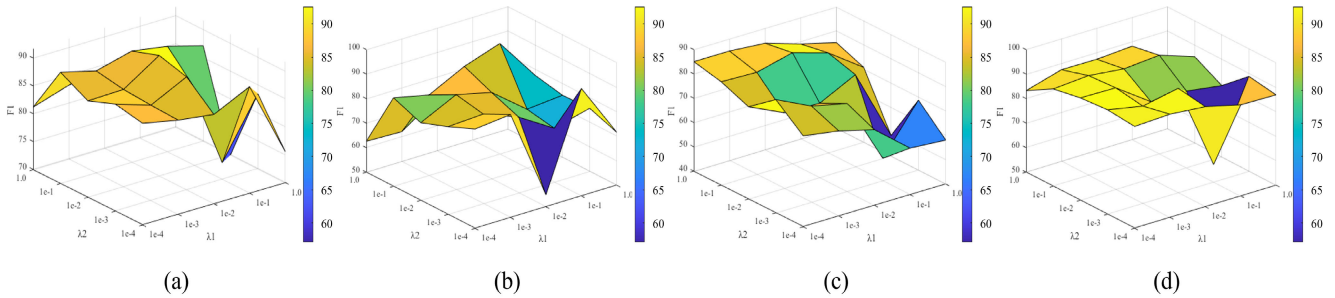


Fig. 9. $F1$ of TGDAnet with different hyperparameters λ_1 and λ_2 on four tasks. (a) Bay→Santa. (b) Santa→Bay. (c) China→USA. (d) USA→China.

In the initial phase, the model has not yet learned sufficiently discriminative cross-domain representations, leading to noisy and semantically unreliable pseudolabels. However, their influence on parameter optimization is limited and does not dominate the overall training process. As training proceeds, the model progressively improves its feature representation and discrimination capability by leveraging supervision from the SD, which results in increasingly reliable pseudolabels. This mutual reinforcement between pseudolabel quality and model learning forms a positive feedback loop, ultimately driving the performance to continuously improve and converge to a stable state.

D. Parameter Analysis and Discussions

We obtain the optimal results for TGDAnet by tuning the hyperparameters λ_1 and λ_2. These parameters are searched within the range of 0.0001–1.0 with increments of 10 to identify the optimal combination that maximizes TGDAnet's performance. Fig. 8 shows the 3-D figures of IoU accuracy after tuning the hyperparameters λ_1 and λ_2 across four tasks, illustrating their effect on IoU. Meanwhile, Fig. 9 presents the 3-D figures of $F1$ for the same tasks, highlighting the impact of these two hyperparameters on $F1$. Figs. 8 and 9 clearly show that the trends in IoU and $F1$ are similar across the four tasks with tuning the hyperparameters λ_1 and λ_2. However, the minimum accuracy of $F1$ is significantly higher than that of IoU.

From Figs. 8(a) and 9(a), it can be seen that in the "Bay→Santa" task, IoU and $F1$ initially increase as the values of λ_1 and λ_2 rise, but eventually begin to decrease as these hyperparameters continue to increase. Therefore, when $\lambda_1 = 0.01$ and $\lambda_2 = 0.01$, both IoU and $F1$ are maximized. As shown in Figs. 8(b) and 9(b), IoU and $F1$ are more influenced by λ_1 and λ_2 in the "Santa→Bay" task. For λ_1, IoU and $F1$ initially increase, then suddenly drop, rise again, and drop once more. For λ_2, IoU and $F1$ first decrease slowly, followed by a sharp decline, a subsequent rise, and then a slow decline again. Therefore, when $\lambda_1 = 0.1$ and $\lambda_2 = 0.1$, both IoU and $F1$ are maximized. As shown in Figs. 8(c) and 9(c), in the "China→USA" task, IoU and $F1$ decrease as λ_1 increases. Meanwhile, IoU and $F1$ initially increase with rising values of λ_2, but eventually start to decrease as λ_2 continues to increase. Therefore, when $\lambda_1 = 0.0001$ and $\lambda_2 = 0.001$, both IoU and $F1$ are maximized. From Figs. 8(d) and 9(d), it can be seen that in the "USA→China" task, IoU and $F1$ are less affected by λ_1, showing only small changes in the early stages and a decline in the later stages. In contrast, IoU and $F1$ are more influenced by λ_2, with small changes initially, a sudden drop in the middle stage, and an upward trend later on. Therefore, when $\lambda_1 = 0.0001$ and $\lambda_2 = 0.001$, both IoU and $F1$ are maximized.

E. Ablation Analysis and Discussions

TGDAnet primarily consists of an SS-Transformer architecture (*backbone*), vision features(*vision*), text features (*text*), domain adaptation (*da*), global losses ($\mathcal{L}_{global}$), and local losses ($\mathcal{L}_{local}$). We design a series of ablation experiments to effectively assess the specific contributions of these components to the performance of TGDAnet. Table II shows the impact of enabling or disabling each component on IoU and $F1$ across four tasks. The optimal results are highlighted in bold, while the suboptimal results are underlined.

As shown in Table II, compared with the *backbone*, using only the *text* leads to a significant degradation in CD accuracy.

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TABLE II
IoU AND $F1$ FOR THE FOUR TASKS WITH DIFFERENT COMPONENTS

Components					Tasks							
backbone	*vision*	*text*	*da*		Bay→Santa		Santa→Bay		China→USA		USA→China	
			$\mathcal{L}_{local}$	$\mathcal{L}_{global}$	IoU	F1	IoU	F1	IoU	F1	IoU	F1
✓					67.46	80.57	55.02	70.99	74.93	85.67	70.61	82.78
✓		✓			37.32	54.36	18.49	31.21	19.10	32.08	2.90	5.64
✓		✓	✓	✓	39.33	56.46	20.41	33.15	36.50	53.48	8.30	15.32
✓	✓		✓	✓	80.98	89.49	69.27	81.85	72.65	84.16	85.87	92.40
✓	✓	✓		✓	86.67	92.86	90.38	94.95	73.31	84.60	**87.08**	**93.09**
✓	✓	✓	✓		**86.97**	**93.03**	87.14	93.13	74.53	85.41	86.05	92.50
✓	✓	✓	✓	✓	84.53	91.62	**90.52**	**95.03**	**76.38**	**86.61**	86.25	92.62

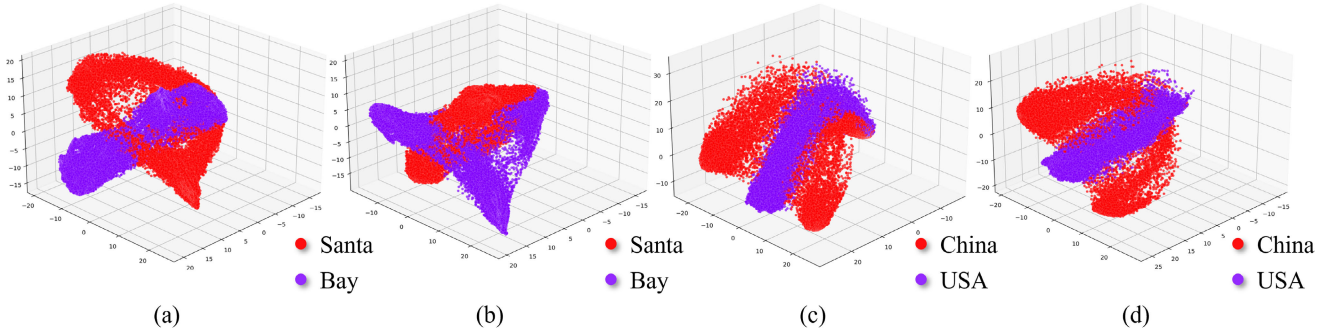


Fig. 10. Visualization of vision and vision–text features across four scenes. (a) Visual features extracted from the Santa and Bay scenes. (b) Vision–text features extracted from the Santa and Bay scenes. (c) Visual features extracted from the USA and China scenes. (d) Vision–text features extracted from the USA and China scenes.

However, the introduction of *da* partially alleviates this performance drop. Specifically, the *da* module achieves an accuracy improvement of approximately 10% over the *backbone* across three tasks, although negative transfer is observed in the "China→USA" scenario. Further improvements are obtained by incorporating textual features into the *da* framework and applying either $\mathcal{L}_{global}$ or $\mathcal{L}_{local}$. During the experiments, we observe that rivers in the USA dataset are frequently misclassified as changed regions, which we believe is the primary cause of the negative transfer in the "China→USA" task. In this context, *text* plays a crucial auxiliary role in disambiguating such complex semantic changes. In the "Bay→Santa" task, the joint application of $\mathcal{L}_{global}$ and $\mathcal{L}_{local}$ slightly degrades performance, likely due to the relatively small interdomain gap, where overly strong constraints hinder further accuracy gains. Although the combined strategy does not achieve optimal performance across all the four tasks, it attains the best results in two tasks and suboptimal results in one, which is still superior to other configurations. Moreover, Table II indicates that $\mathcal{L}_{global}$ and $\mathcal{L}_{local}$ provide complementary benefits, and neither loss alone is sufficient to consistently improve CD accuracy. Overall, these results demonstrate that TGDAnet establishes a robust and effective framework for HSIs-CD.

Fig. 10 shows the visualization of vision and vision–text features in the four scenes. A comparative analysis of Fig. 10(a) and 10(b) indicates that the boundary between Santa and Bay is more distinct in Fig. 10(a), whereas it appears relatively blurred in Fig. 10(b). Similarly, a comparison of Fig. 10(c) and 10(d) shows that the boundary between the USA and China is clearer in Fig. 10(c), while it becomes less defined in Fig. 10(d). As can be seen from Fig. 10, there are obvious distributional differences in vision features between different scenes, and it is often difficult to obtain ideal performance by directly transferring the knowledge learned in the SD to the TD. However, after fusing vision and text features, the difference in the distribution of features between the SD and TD is reduced, at which point domain adaptation is able to perform more effectively. As can be seen in Fig. 10, it must be honestly said that the present method still fails to fully realize the complete alignment of feature distributions between different scenes.

F. CD Results

Table III presents the quantitative results of TGDAnet and the comparison methods for IoU, $F1$, Prec, and Recall, with the optimal results highlighted in bold and the suboptimal results underlined. In addition, Figs. 11–14 display the qualitative results of these methods across the four tasks.

1) *"Bay→Santa" Task*: From Table III, it can be seen that TGDAnet demonstrates excellent CD capability, significantly outperforming other methods in both IoU and $F1$. Compared with the suboptimal method, TGDAnet improves IoU by 2.88% and $F1$ by 1.72%. In addition,

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TABLE III
PREDICTED QUANTITATIVE RESULTS OF DIFFERENT METHODS ON FOUR TASKS

Methods	Bay→Santa				Santa→Bay				China→USA				USA→China			
	IoU	F1	Prec	Recall	IoU	F1	Prec	Recall	IoU	F1	Prec	Recall	IoU	F1	Prec	Recall
DAAN	70.37	82.61	0.7252	0.9595	58.58	73.88	0.9154	0.6193	56.51	72.21	0.7369	0.7080	68.33	81.19	0.8348	0.7901
DDAN	74.69	85.51	0.7631	**0.9724**	70.09	82.42	**0.9885**	0.7067	52.90	69.20	0.5944	0.8279	82.24	90.25	0.9071	0.8980
TSTnet	78.66	88.06	0.8275	0.9409	86.08	92.52	0.9623	0.8908	57.13	72.72	0.7124	0.7426	72.04	83.75	0.9136	0.7731
CMJAN	77.69	87.45	0.8886	0.8607	88.98	94.17	0.9480	**0.9354**	58.21	73.59	**0.9482**	0.6012	79.54	88.60	0.8537	0.8209
CACL	78.69	88.07	**0.9250**	0.8405	88.21	93.74	0.9681	0.9085	67.28	80.44	0.7424	**0.8776**	78.94	88.23	**0.9371**	0.8336
S^{4}DL	81.65	89.90	0.8361	0.9721	83.04	90.73	0.9780	0.8462	67.18	80.37	0.8353	0.7744	84.54	91.62	0.9205	0.9120
TGDAnet	**84.53**	**91.62**	0.8846	0.9501	**90.52**	**95.03**	0.9710	0.9304	**76.38**	**86.61**	0.8845	0.8484	**86.25**	**92.62**	0.8901	**0.9653**

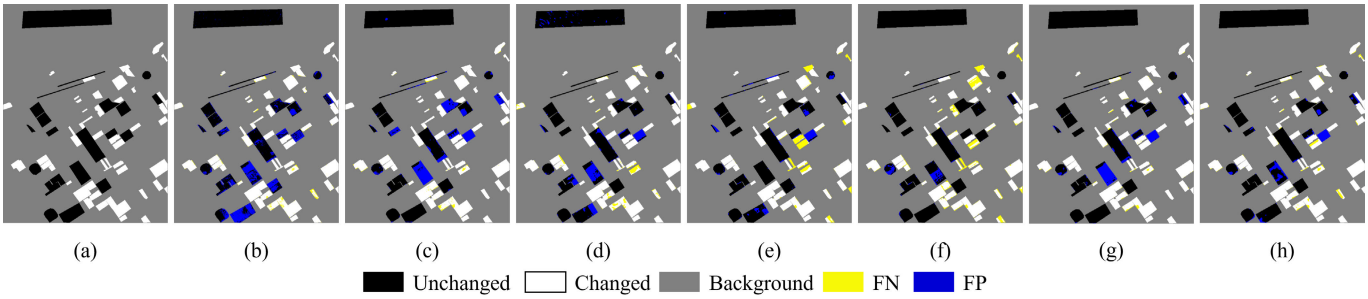


Fig. 11. Predicted results for Santa in the "Bay→Santa" task. (a) GT. (b) DAAN. (c) DDAN. (d) TSTnet. (e) CMJAN. (f) CACL. (g) S^{4}DL. (h) TGDAnet.

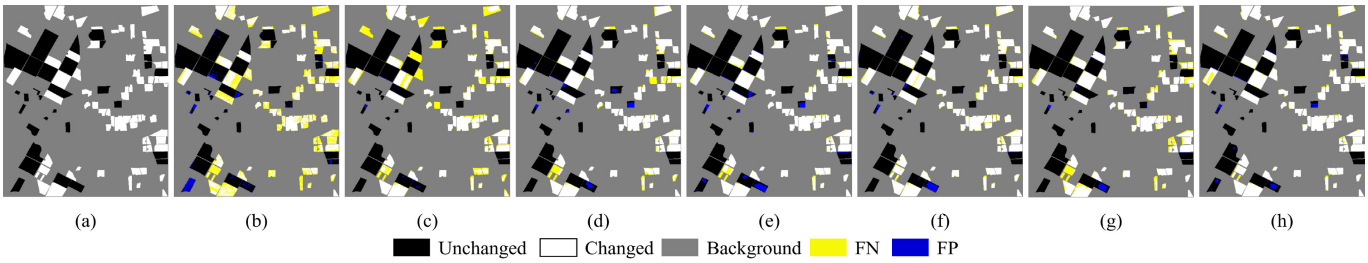


Fig. 12. Predicted results for Bay in the "Santa→Bay" task. (a) GT. (b) DAAN. (c) DDAN. (d) TSTnet. (e) CMJAN. (f) CACL. (g) S^{4}DL. (h) TGDAnet.

TGDAnet is only 0.0404 lower than the optimal method in Prec and 0.0223 lower in Recall.

Fig. 11 illustrates the qualitative evaluation results of several methods for the "Bay→Santa" task. As shown, the blues and yellows for DAAN, DDAN, and TSTnet are significantly higher than those for TGDAnet when compared with traditional methods. Among the novel methods, CACL shows significantly lower blues but higher yellows than TGDAnet, while S^{4}DL exhibits much higher blues and yellows than TGDAnet. In comparison to CMJAN, TGDAnet also outperforms in blues, though CMJAN has higher yellows.

2) *"Santa→Bay" Task*: From Table III, TGDAnet improves IoU by 1.54% and $F1$ by 0.86%, but exhibits suboptimal results for Prec and Recall, being 0.0175 lower than the optimal method in Prec and 0.0050 lower in Recall.

Fig. 12 illustrates the qualitative evaluation results for the "Santa→Bay" task from several methods. As shown, DAAN, DDAN, and TSTnet have significantly higher yellows than TGDAnet, accompanied by notably higher miss detection rates. Among the novel methods, CACL and S^{4}DL show similar blues to TGDAnet but with higher yellows. In comparison, the blues and yellows of CMJAN are visually similar to those of TGDAnet.

3) *"China→USA" Task*: From Table III, TGDAnet also shows improvements: it enhances IoU by 9.1% and $F1$ by 6.17%, though it is 0.0637 lower than the optimal method in Prec and 0.0292 lower in Recall.

Fig. 13 illustrates the qualitative evaluation results of the "China→USA" task from several methods. As shown, the blues and yellows of DAAN, DDAN, and TSTnet are significantly higher than those of TGDAnet when compared with traditional methods. It is observed that DAAN, DDAN, and TSTnet fail to detect changes along the riverbank. Among the novel approaches, CACL has significantly lower blues than TGDAnet, with yellows significantly higher, but it shows substantial detection errors despite observing riverbank changes. S^{4}DL shows higher blues and yellows than TGDAnet and fails to detect changes along the riverbank. In comparison, the

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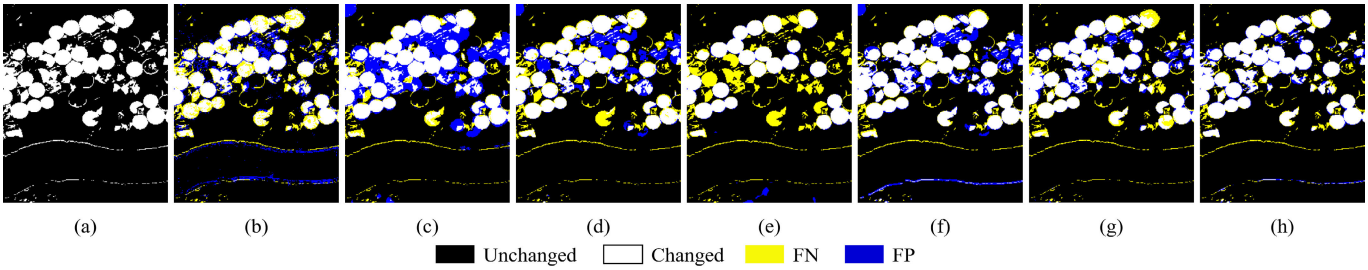


Fig. 13. Predicted results for USA in the "China→USA" task. (a) GT. (b) DAAN. (c) DDAN. (d) TSTnet. (e) CMJAN. (f) CACL. (g) S^{4}DL. (h) TGDAnet.

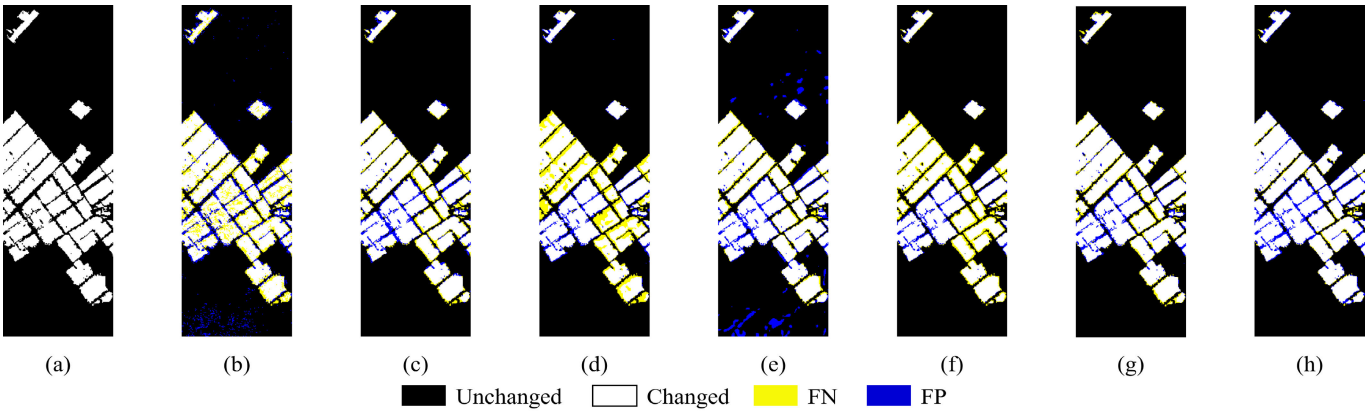


Fig. 14. Predicted results for China in the "USA→China" task. (a) GT. (b) DAAN. (c) DDAN. (d) TSTnet. (e) CMJAN. (f) CACL. (g) S^{4}DL. (h) TGDAnet.

TABLE IV
TIME CONSUMPTION (SECONDS) OF METHODS ON FOUR TASKS

Task	DAAN	DDAN	TSTnet	CMJAN	CACL	S^{4}DL	TGDAnet
Bay→Santa	433.11	980.17	707.59	454.30	300.91	357.39	675.79
Santa→Bay	962.72	1625.44	1757.93	454.30	757.47	797.44	1091.57
China→USA	240.18	723.30	473.60	278.87	328.79	175.72	475.18
USA→China	321.83	879.99	628.84	278.87	389.26	248.97	542.58

blues and yellows of CMJAN are visually similar to those of TGDAnet. Compared with CMJAN, it has much lower blues than TGDAnet, and while the yellows are higher, the miss detection rate is notably high.

4) *"USA→China" Task*: As shown in Table III, TGDAnet demonstrates superior CD capabilities in terms of IoU, $F1$, and Recall, significantly outperforming other methods. Compared with the suboptimal method, TGDAnet improves IoU by 1.71%, $F1$ by 1.00%, and recall by 0.0533. In addition, TGDAnet is 0.047 lower than the optimal method in terms of Prec.

Fig. 14 illustrates the qualitative evaluation results for the "USA→China" task from several methods. As shown, DAAN, DDAN, and TSTnet have significantly higher blues and yellows than TGDAnet, along with substantially higher false and missed detection rates compared with traditional methods. Among the novel methods, CACL and S^{4}DL exhibit similar blues to TGDAnet, but their yellows are higher, and they also show a significantly higher false detection rate. In comparison, CMJAN has similar yellows to TGDAnet but significantly higher blues and an even higher false detection rate.

5) *Time Complexity*: We conduct a comprehensive empirical test of TGDAnet's time complexity on a device equipped with an NVIDIA GeForce RTX 3090. Following this, we perform a detailed comparative analysis of the performance of TGDAnet against other methods, with the results summarized in Table IV. Further analysis, along with the experimental procedure, reveals that the primary time consumption in TGDAnet comes from the text encoding process, while feature extraction by the transformer also contributes significantly to the time overhead. However, given the substantial improvement in CD performance of TGDAnet, we consider this time investment to be reasonable and worthwhile.

V. CONCLUSION

In this article, we propose a TGDAnet for hyperspectral binary CD. The method begins with preprocessing the source

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and TDs to align the spectral dimensions of the images across different domains. Next, deep features from both the domains are extracted using an SS-Transformer. Then, vision and text features are fused multiple times based on the text guide. Finally, contrastive learning is used to construct global and local domain adaptation losses, which help reduce the differences in feature distribution between the source and TDs, enabling the network to better detect subtle changes. In addition, we provide a comprehensive evaluation of our method through ablation experiments, parametric analysis, and comparisons with other major methods. These results validate the effectiveness and significance of our proposed method for hyperspectral CD. In future studies, we plan to further explore the spectral advantages of HSIs and optimize the complexity of the model to enable faster and more efficient CD.

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The numerical calculations in this article have been done on the supercomputing system in the Supercomputing Center of Wuhan University.

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